

Rahul Rakesh

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EDUCATION

University of Western Ontario

London, Ontario

B.Sc. Honours Specialization in Computer Science (CO-OP)

2023 – 2028

- Relevant Coursework: Data Structures & Algorithms, Database, Computer Architecture, Operating Systems

EXPERIENCE

Machine Learning Engineer

Jan. 2026 – Present

Fly Rank

Remote

- Conducted machine learning research on signal analysis by processing and analyzing datasets to identify meaningful patterns and evaluate model performance
- Developed Python-based data analysis workflows and visualizations to support experimentation, interpret results, and improve research efficiency
- Collaborated with mentors to document research methodologies, experimental findings, and technical insights for an ongoing signal analysis research paper

Freelance Web Developer

Sept. 2024 – Present

Self-Employed

Remote

- Delivered production-grade full-stack applications for 5+ clients using React, Node.js, Express.js, and PostgreSQL with RESTful API design
- Improved website performance with 50% faster load times and 40% higher engagement through code optimization, lazy loading, and SEO best practices
- Led projects end-to-end: requirements gathering, system design, CI/CD pipelines, and deployment to AWS and Vercel

PROJECTS

Fake Out | *React, TypeScript, Node.js, Socket.IO, Docker, Fly.io*

- Built a real-time multiplayer game with sub-100ms latency and 99.5% uptime using React, TypeScript, Node.js, and Socket.IO
- Designed authoritative server-side game state with room-based events and heartbeat monitoring, supporting 10 concurrent players without desynchronization
- Deployed containerized services via Docker on Fly.io with CI/CD pipelines, handling 1,000+ game sessions

Schedula | *Azure Functions, Azure SQL, JWT, REST API*

- Built a serverless booking API with sub-200ms response times using Azure Functions, Azure SQL, and JWT authentication
- Prevented race conditions and double-bookings through transaction-safe scheduling, SQL isolation levels, and idempotent design
- Modeled a normalized relational schema with indexed foreign keys to support efficient multi-user concurrent queries under load

VitaLink | *React Native, Django REST, PostgreSQL | TerraHacks 2025*

- Reduced patient intake time by 30% with a full-stack healthcare platform built in React Native and Django REST Framework
- Integrated third-party APIs into backend service pipelines, processing structured outputs for downstream features and form pre-filling
- Implemented role-based access control with token authentication to enforce data privacy across patient, provider, and admin tiers

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Python, Java, JavaScript, TypeScript, C/C++, SQL

Web & Backend: React, Node.js, Express.js, Django REST Framework, REST APIs, Socket.IO, React Native

Databases & Caching: PostgreSQL, MongoDB, SQLite, Azure SQL (schema design, indexing, two-phase locking, query optimization)

Cloud & DevOps: AWS (EC2, S3, Lambda), Azure Functions, Docker, CI/CD (GitHub Actions), Vercel, Fly.io

Certifications: Anthropic (Claude with the Anthropic API, Claude 101, AI Fluency) | Udemy (Intro to Cloud Computing on AWS)